# Micro I Notes

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## Preference

### Def "binary relation"

A binary relation $R$ on $X$ is a subset of $X \times X$.

The ordered pair $(x, y) \in R$ means $x$ is (weakly) preferred to $y$.

#### Remark about notation

We use $\succsim$ to denote $R$ but all these methods are the same thing:

$$(x, y) \iff xRy \iff x \succsim y$$

### Def "attributes of binary relations"

A binary relation $\succsim$ is

1. complete if for all $x, y \in X$ either $x \succsim y$ or $y \succsim x$
2. transitive if for all $x, y, z \in X, [x \succsim y, y \succsim z] \Rightarrow x \succsim y$
3. reflexive if for all $x \in X, x \succsim x$
4. symmetric if for all $x, y \in X, x \succsim y \iff y \succsim x$
5. irreflexive if for all $x \in X, x \not\succsim x$
6. asymmetric if for all $x, y \in X, x \succsim y \Rightarrow y \not\succsim x$

### Def "rational"

A binary relation $\succsim$ is *rational* if it is complete and transitive

## Choice

### Def "choice function"

A function $c : \mathcal{B} \to \mathcal{P}(X)$ is called a *choice function* if for all $A \in \mathcal{B}, c(A) \subseteq A$.

1. If $c$ is a choice function, then $c(A) \neq \emptyset, \forall A \in \mathcal{B}$
2. $|c(A)| \geq 1$

### Def "choice structure"

The pair $(\mathcal{B}, c)$ is called a *choice structure*.

### Def "WARP"

Weak Axiom of Revealed Preference. The following are equivalent for all $A, B \in \mathcal{B}$:

- $c(A) \cap B \neq \emptyset \Rightarrow c(B) \cap A \subseteq c(A)$
- $[c(A) \cap B \neq \emptyset \text{ AND } c(B) \cap A \neq \emptyset] \Rightarrow [c(A) \cap B \subseteq c(B) \text{ AND } c(B) \cap A \subseteq c(A)]$
- $x, y \in A \cap B, x \in c(A), y \in c(B) \Rightarrow x \in c(B), y \in c(A)$
- Sen's $\alpha$ and $\beta$ both hold

### Def Sen's $\alpha$ and $\beta$

For all $A, B \in \mathcal{B}$:

1. Sen's $\alpha$: If $x \in B \subseteq A$ and $x \in c(A)$ then $x \in c(B)$
2. Sen's $\beta$: If $x, y \in c(B), B \subseteq A$ and $y \in c(A)$ then $x \in c(A)$

### Prop 1 "WARP $\iff$ $\alpha$ and $\beta$"

### Term "single-valued"

We say $c$ is single-valued if $\forall A \in \mathcal{B}, |c(A)| = 1$

### Prop 2 "$\alpha$ and single-valued $\Rightarrow$ WARP"

If $(\mathcal{P}(X), c)$ satisfies $\alpha$ and $c$ is single-valued, then it satisfies WARP.

## Choice and Preference

### Def "$c_{\succsim}$"

For any binary relation $\succsim$ define

$$c_{\succsim}(A) = \{x \in A : x \succsim y, \forall y \in A\}, \forall A \in \mathcal{P}(X)$$

### Def "$\succsim_c$"

For a choice structure $(\mathcal{P}(X), c)$ define a binary relation $\succsim_c$ such that

$$x \succsim_c y \iff x \in c(\{x, y\})$$

### Prop "satisfies WARP $\iff$ $\succsim$ rational

If $\succsim$ is a binary relation on a finite set $X$ then $(\mathcal{P}(X), c_{\succsim})$ is a choice structure that satisfies WARP if and only if $\succsim$ is rational.

### Prop "WARP $\Rightarrow c = c_{\succsim}$"

If $(\mathcal{P}, c)$ is a choice structure that satisfies WARP then there exists a unique rational binary relation $\succsim$ such that $c = c_{\succsim}$ (i.e., for all $A \in \mathcal{P}, c(A) = c_{\succsim}(A)$).

## Utility

### Def "utility function"

A function $U : X \to \mathbb{R}$ represents $\succsim$ if, for all $x, y \in X$

$$x \succsim y \iff U(x) \geq U(y).$$

### Prop "$U$ represents $\succsim \Rightarrow$ rational"

If $U : X \to \mathbb{R}$ represents $\succsim$, then $\succsim$ is rational.

Note: this proposition does NOT require finiteness!

### Prop "$X$ finite + $\succsim$ rational $\Rightarrow \exists U$

If $X$ is finite and $\succsim$ is rational then there exists $U : X \to \mathbb{R}$ that represents $\succsim$.

### Prop "$V \iff \phi$ and $\phi$ strictly increasing"

Suppose $U : X \to \mathbb{R}$ represents $\succsim$. Then $V : X \to \mathbb{R}$ represents $\succsim$ if and only if there exists $\phi : U(X) \to V(X)$ such that $V = \phi \circ U$ and $\phi$ is strictly increasing.

## Expected Utility

### Def "lottery"

Suppose $X$ is a non-empty and finite set of prizes (consequences). Then a *lottery*, $p$ is a probability distribution over $X$:

$$p : X \to [0, 1], \text{ such that } \sum_{x \in X} p(x) = 1.$$

We denote $\mathcal{L}$ as the set of all lotteries.

#### Term "degenerate lottery"

If $p(x) = 1$ for some $x \in X$ then we write $\delta_x$.

**Remark: $\mathcal{L}$ is the set of alternatives, and unlike $X$ is infinite!**

### Def "expected utility function/utility index"

$U : \mathcal{L} \to \mathbb{R}$ is an expected utility function if there exists $u : X \to \mathbb{R}$ such that $U(p) = \sum_{x \in X} u(x)p(x), \forall p \in \mathcal{L}$.

Called a "von Neumann–Morgenstern utility index" or a "Bernoulli index."

### Def "Mixture operation"

For all $p, q \in \mathcal{L}, \alpha \in [0, 1]$:

- let $\alpha p + (1 - \alpha)q$ denote a lottery $r \in \mathcal{L}$ such that for all $x \in X, r(x) = \alpha p(x) + (1 - \alpha)q(x)$.

### Def "expected utility representation"

$\succsim$ on $\mathcal{L}$ has an expected utility representation if there exists $u : X \to \mathbb{R}$ such that for $p, q \in \mathcal{L}$

$$p \succsim q \iff U(p) = \sum_{x \in X} p(x)u(x) \geq U(q) = \sum_{x \in X} q(x)u(x)$$

### Def "linear function"

A function $U : \mathcal{L} \to \mathbb{R}$ is *linear* if for all $\alpha \in [0, 1]$ and $p, q \in \mathcal{L}$:

$$U(\alpha p + (1 - \alpha)q = \alpha U(p) + (1 - \alpha)U(q).$$

## Lemma "linear $\iff$ $U$ is an expected utility function"

A function $U : \mathcal{L} \to \mathbb{R}$ is *linear* if and only if $U$ is an expected utility function.

## Def "vNM Axioms"

The von Neumann–Morgenstern axioms are:

- A1. (Rationality): $\succsim$ is complete and transitive.
- A2. (Independence): For all $p, q, r \in \mathcal{L}, \alpha \in (0,1)$ we have $[p \succsim q] \Rightarrow [\alpha p + (1-\alpha)r \succsim \alpha q + (1-\alpha)r]$.
- A3. (Continuity): For all $p, q, r \in \mathcal{L}$, if $p \succ q \succ r$ then there exists $\alpha, \beta \in (0,1)$ such that $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.
- We don't use A2* in this class, but some literature uses it:
  - A2*. (Independence, strengthened): $[p \succsim q] \iff [\alpha p + (1-\alpha)r \succsim \alpha q + (1-\alpha)r]$

## Lemma "applying vNM Axioms"

Suppose $\succsim$ satisfies A1, A2, and A3 of vNM axioms.

(i) $\forall \alpha \in (0,1), p, q \in \mathcal{L}$:
  (a) $p \succ q \Rightarrow p \succ \alpha p + (1-\alpha)q \succ q$
  (b) $p \sim q \Rightarrow p \sim \alpha p + (1-\alpha)q \sim q$
(ii) $p \succ q, 1 \geq \alpha > \beta \geq 0 \Rightarrow \alpha p + (1-\alpha)q \succ \beta p + (1-\beta)q$
(iii) $p \sim q, \alpha \in (0,1) \Rightarrow \alpha p + (1-\alpha)r \sim \alpha q + (1-\alpha)r$
(iv) $p \succsim q, r \succsim s, \alpha \in (0,1) \Rightarrow \alpha p + (1-\alpha)r \succsim \alpha q + (1-\alpha)s$

## Thm "A1-A3 $\iff$ EU representation"

Suppose $X$ is finite and $\succsim$ is a binary relation on $\mathcal{L}$. Then

$$[\succsim \text{ satisfies A1, A2, and A3 }] \iff \succsim \text{ has an Expected Utility representation.}$$

### Thm "$V$ linear representation $\iff$ $V(p) = aU(p) + b$"

Suppose $U$ is a linear representation of $\succsim$, and $V : \mathcal{L} \to \mathbb{R}$ is a linear function. Then

$$[V \text{ is a linear representation of } \succsim] \iff [\exists a > 0, b \in \mathbb{R} \text{ such that } V(p) = aU(p) + b, \forall p \in \mathcal{L}]$$

### Def "simple lottery"

$P : X \to \mathbb{R}$ is a *simple lottery* if

1. $\{x \in X : p(x) > 0\}$ is finite
2. $\displaystyle\sum_{x \in X : p(x) > 0} p(x) = 1$

### Def "simple improvement"

For all (simple) lotteries $p, q \in \mathcal{L}$, $p$ is a simple improvement of $q$ if there exists $\lambda \in (0,1), m, m' \in X, m > m; , r \in \mathcal{L}$ such that

$$p = \lambda \delta_m + (1-\lambda)r \text{ and } q = \lambda \delta'_m + (1-\lambda)r.$$